RESEARCH PROJECT PLAN
Seminar Project · HSE University · 2026

Detection of AI-Generated Text

Classical ML · Neural Networks · Topological Data Analysis

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Type	Seminar Research Project
Status	Planning / In Progress
Year	2026

1. RATIONALE AND MOTIVATION

The widespread adoption of large language models — GPT-4, LLaMA, Gemini — creates a growing threat of misuse: academic plagiarism, disinformation, and fabricated content. Automated detection of AI-generated text is becoming a critical research challenge.

This project compares classical ML approaches with neural network methods for binary classification of human vs AI-generated text. It also explores Topological Data Analysis (TDA) as an alternative approach grounded in the structural properties of language — capable of 93–97% accuracy without any neural network.

Core Research Questions

- Which approach performs best: classical ML (TF-IDF + SVM) or neural (fastText+LSTM, TinyBERT)?
- How does the topological complexity of AI-generated text differ from human-written text?
- Is the 'perforation metric' from the Hidden Holes paper applicable to AI detection?
- Which RNN architecture better captures AI text patterns — GRU or LSTM?
- Can we generate soft labels — a confidence score rather than a hard binary decision?

2. THEORETICAL FOUNDATIONS

2.1 "Hidden Holes" — Topological Analysis of Text

The paper (Kushnareva et al.) demonstrates that human-written texts exhibit higher topological complexity than AI-generated ones. This is measured using persistent homology and Vietoris-Rips complexes built on high-dimensional text embeddings.

- Perforation metric: measures the number and persistence of topological holes (Betti numbers β_1) in the embedding space

- Key finding: LSTM-generated text >> Transformer-generated text in topological complexity, explaining why LSTM outputs are harder to distinguish from human text
- Practical application: compute the perforation metric as an additional feature for the classifier

2.2 "Spot the Bot" — Gromov's Method (HSE)

Gromov treats language as a self-organised critical system — analogous to physical systems at a phase transition boundary. AI-generated text violates this criticality, which manifests as measurable statistical anomalies.

- Hausdorff dimension: fractal characteristic of text; AI texts systematically differ from human texts on this measure
- Entropy-complexity plane: AI texts cluster in a specific region of this 2D space
- TDA without neural networks: method achieves 93–97% accuracy using only topological features
- Diagnostic potential: detects stylometric deviations and bot authorship

2.3 Supervisor's Recommendations

The HSE supervisor provided specific guidance on developing this project:

- User scenarios: describe concrete use cases — student essays, news articles, product reviews
- Dataset quality: pay close attention to training data quality — garbage in, garbage out
- Statistical and topological methods: apply TDA alongside neural approaches
- Baseline: fastText + LSTM — recommended as a strong baseline that captures local context
- Architecture comparison: GRU vs LSTM — run a controlled experiment
- Advanced model: try TinyBERT — a distilled BERT variant
- Soft labels: design a confidence metric instead of hard binary classification
- TF-IDF: logically cannot yield high results — include as lower bound to confirm

3. USER SCENARIOS

As directed by the supervisor, the specific use cases and user profiles must be clearly defined.

Scenario A: Academic Settings

- User: lecturer, teaching assistant, anti-plagiarism platform
- Input: student essay, coursework, or seminar paper
- Output: AI-generation probability + highlighted high-confidence passages
- Requirement: high recall (false alarm preferable to missing AI text)

Scenario B: Media & Fact-Checking

- User: editor, fact-checker, publishing platform
- Input: news article, press release, social media post
- Output: credibility score + source classification (human / AI / mixed)
- Requirement: high precision, explainability of the decision

Scenario C: Review Platforms

- User: marketplace or review service (e-commerce)
- Input: product or service review
- Output: 'suspicious review' flag + confidence score
- Requirement: fast inference, batch-processing capability

4. DATA

4.1 Dataset Sources

- HC3 (Human ChatGPT Comparison Corpus) — question-answer pairs: human and ChatGPT
- TuringBench — benchmark for AI-generated text detection
- RAID Dataset — multi-source dataset from diverse LLMs
- AuTextification (IberLEF 2023) — multilingual, includes Spanish
- Reddit / Wikipedia aggregates — human-written reference text

4.2 Quality Requirements

The supervisor explicitly emphasised: dataset quality is critical — garbage in, garbage out.

- Class balance: ~50/50 human vs AI (or class weighting)
- Domain diversity: academic texts, news, forums, reviews
- Model diversity: GPT-3.5, GPT-4, LLaMA, Claude — multiple 'AI authors'
- Remove duplicates, noisy records, and unlabelled samples
- Stratified split: train / validation / test (70 / 15 / 15)

4.3 Preprocessing

- Tokenisation: standard (NLTK / SpaCy) for ML; subword BPE for neural models
- Remove HTML, special characters; normalise whitespace
- For TDA: build embeddings → Vietoris-Rips complexes → compute Betti numbers

5. MODELS AND ARCHITECTURES

The project involves a comparative study of five approaches — from a simple lower-bound baseline to topological methods:

Model	Type	Complexity	Rationale
TF-IDF + SVM	Classical ML (baseline)	Low	Standard lower bound; logically won't yield high results (supervisor's note) — included to confirm
fastText + LSTM	Hybrid (embeddings + sequence)	Medium	Recommended by supervisor as strong baseline — captures local context and morphology
GRU vs LSTM	RNN architecture comparison	Medium	Experiment: which recurrent cell better captures AI text patterns?
TinyBERT	Transformer (distilled)	High	Contextual embeddings; expected F1 leader
TDA Pipeline	Topological analysis	Experimental	Perforation metric + Betti numbers (Hidden Holes paper) + Hausdorff dimension (Gromov)

5.1 fastText + LSTM — Primary Baseline

Recommended by the supervisor as a strong baseline. fastText provides high-quality word embeddings with morphological awareness (important for Russian-language texts). LSTM captures sequential dependencies and local context — key for detecting AI patterns.

- Architecture: fastText embeddings (dim 300) → LSTM (hidden 256) → Dropout → Dense(2)
- Training: Adam optimizer, lr=1e-3, batch_size=64, early stopping
- Soft labels: sigmoid output + threshold tuning for confidence score

5.2 GRU vs LSTM — Comparative Experiment

Goal: determine which recurrent architecture better models AI-text patterns.

- GRU: fewer parameters, faster training, sometimes better on short texts
- LSTM: retains longer-range dependencies — preferable for longer documents
- Experiment: identical architecture, same data — only the recurrent cell differs
- Metrics: F1, AUC, training time, parameter count

5.3 TinyBERT — Advanced Approach

Distilled version of BERT — 7.5x smaller and 9x faster while retaining ~97% of BERT's quality. Contextual embeddings are the expected F1 leader.

- Fine-tuning: huggingface/tiny-bert-for-sequence-classification
- Batch size: 16–32, lr=2e-5, 3–5 epochs, linear warmup
- Enables zero-shot and few-shot detection experiments

6. SOFT LABELS AND CONFIDENCE SCORING

As directed by the supervisor: design a confidence metric instead of hard binary classification. AI texts frequently contain mixed content — human-edited AI output. A soft score is more informative than a binary label.

Proposed Approaches

- Model Confidence Score: softmax probability ($P(AI) \in [0, 1]$) — simple baseline
- Ensemble Agreement: fraction of ensemble models predicting AI — interpretable metric
- Sentence-Level Scoring: score each sentence individually → aggregate (mean, max) → mixed-content detection
- Topological Confidence: normalised perforation metric as a measure of 'AI-likeness'
- Calibrated Probability: temperature scaling to calibrate neural network probabilities

Final confidence score: weighted combination — model confidence (0.5) + topological score (0.3) + linguistic features (0.2).

7. EVALUATION METRICS

Metric	Description	Why It Matters
Accuracy	Fraction of correct predictions	Primary performance indicator
F1-Score	Harmonic mean of precision and recall	Handles class imbalance
ROC-AUC	Area under the ROC curve	Threshold-independent quality measure
Confidence	Model certainty per sample (0–1)	Soft labels — key research goal

Metric	Description	Why It Matters
Score		
Topological Complexity	Number of holes (Betti numbers) in data	Distinguishes human vs AI structure
Hausdorff Dimension	Fractal dimension of text	Self-organised criticality criterion (Gromov)

8. PHASED WORK PLAN

Phase 1: Data Collection & EDA · 2–3 weeks

- Collect and merge datasets (HC3, TuringBench, RAID)
- Analyse class distributions, domains, and text lengths
- Preprocessing: tokenisation, cleaning, normalisation
- Stratified train / val / test split
- Quality audit: deduplication, noise removal

Phase 2: Baseline: TF-IDF + Classical ML · 1–2 weeks

- TF-IDF vectorisation (unigrams + bigrams)
- Train SVM, Logistic Regression, Random Forest
- Evaluate Accuracy, F1, AUC — expected low results
- Document as confirmed lower bound

Phase 3: Neural Baselines: fastText + LSTM/GRU · 2–3 weeks

- Train fastText embeddings (or use pre-trained)
- Build LSTM classifier with dropout and early stopping
- Parallel run: identical architecture with GRU cells
- GRU vs LSTM comparison: F1, AUC, speed, parameters
- Implement confidence score (softmax + threshold tuning)

Phase 4: TinyBERT Fine-tuning · 2 weeks

- Fine-tune TinyBERT on binary classification task
- Experiment with learning rate, batch size, epochs
- Calibration: temperature scaling for probabilities
- Sentence-level scoring for mixed-content detection

Phase 5: Topological Data Analysis (TDA) · 2–3 weeks

- Build embeddings → Vietoris-Rips complexes
- Persistent homology: Betti numbers β_0, β_1
- Perforation metric following the Hidden Holes paper
- Hausdorff dimension + entropy-complexity (Gromov's method)
- TDA as standalone classifier + as additional feature for neural models

Phase 6: Comparison & Final Report · 1–2 weeks

- Summary table of all model results
- Error analysis (confusion matrix, failure case review)
- Implement final ensemble confidence score
- Write-up and results presentation

9. TECHNICAL STACK

Language & Environment	Python 3.11 · Jupyter Notebook · VS Code
ML / DL	scikit-learn · PyTorch · Hugging Face Transformers
NLP	NLTK · SpaCy · fastText · BPE Tokenizer
TDA	Ripser · Gudhi · Persim (persistent homology)
Experiment Tracking	MLflow · Weights & Biases
Data	Pandas · NumPy · HuggingFace datasets
Visualisation	Matplotlib · Seaborn · Plotly
Version Control	Git · GitHub · DVC (data versioning)

10. EXPECTED RESULTS

Model Performance

- TF-IDF + SVM: Accuracy ~70–75%, F1 ~0.68 — confirmed weak baseline
- fastText + LSTM: Accuracy ~85–90%, F1 ~0.87 — strong recommended baseline
- GRU: comparable to LSTM, potentially faster on short texts
- TinyBERT: Accuracy ~92–95%, F1 ~0.93 — expected leader
- TDA (Gromov-style): Accuracy ~88–93% without neural networks — novel contribution

Scientific Contribution

- First comparison of TDA methods (Hidden Holes + Gromov) against neural networks on AI detection
- Design and evaluation of an integrated confidence score based on model ensemble
- Reproduction and validation of the perforation metric on new data
- Practical guidance on approach selection for different deployment scenarios

11. RISKS AND LIMITATIONS

- Data drift: models trained on specific LLMs — newer GPT/Claude versions may evade the detector
- Compute requirements: TinyBERT requires GPU; TDA has high complexity for long texts
- Dataset quality: difficulty verifying that 'human' text was not AI-assisted
- Multilingual scope: most datasets are English; Russian-language texts require separate treatment
- Adversarial robustness: paraphrasing and AI-style masking — system resilience untested

KEY REFERENCES

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[3] Guo et al. "How Close is ChatGPT to Human Experts?" — HC3 dataset, ChatGPT detection baseline.

[4] Jia et al. "TinyBERT: Distilling BERT for Natural Language Understanding" — TinyBERT architecture.

[5] Boyarkov et al. "RAID: A Shared Benchmark for Robust Evaluation of Machine-Generated Text Detectors" — 2023.